Light

Network Protocols & Internet

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AQA GCSE Computer Science — **Network Protocols and the TCP/IP Model (8525 Paper 2)**. This topic covers protocol definitions, the purpose of TCP, IP, HTTP, HTTPS, SMTP and IMAP, the 4-layer TCP/IP model, and network security methods including authentication, encryption, firewalls, and MAC address filtering.

Exam questions typically ask you to **define** what a protocol is, **explain** the purpose of specific protocols, **describe** the TCP/IP layers and their functions, **state** which protocols operate at which layer, and **explain** how security methods protect networks. Questions range from 1-mark definitions to 4-mark explanations.

Why network protocols matter

When you send an email, visit a website, or stream a video, your data travels across networks containing millions of different devices made by thousands of different manufacturers. For all these devices to communicate successfully, they must follow agreed rules — these rules are called **protocols**.

Without protocols, a computer made by one company would not be able to communicate with a server made by another. Protocols ensure that all devices "speak the same language" and handle data in a consistent, predictable way.

- **TCP:** Ensures reliable delivery of data between devices

- **IP:** Addresses and routes data packets across networks
- **HTTP/HTTPS:** Transfers web pages between browsers and servers
- **SMTP/IMAP:** Handles sending and receiving emails

What you must know

This checklist covers the key exam-visible concepts from the AQA specification. You should be able to explain each point clearly.

- **Define** the term network protocol.
- **Explain** the purpose of TCP (Transmission Control Protocol).
- **Explain** the purpose of IP (Internet Protocol).
- **Explain** the purpose of HTTP (Hypertext Transfer Protocol).
- **Explain** the purpose of HTTPS (Hypertext Transfer Protocol Secure).
- **Explain** the purpose of SMTP (Simple Mail Transfer Protocol).
- **Explain** the purpose of IMAP (Internet Message Access Protocol).
- **Describe** the 4-layer TCP/IP model and the function of each layer.
- **State** which protocols operate at which layer of the TCP/IP model.
- **Understand** the need for, and importance of, network security.
- **Explain** how authentication, encryption, firewalls, and MAC address filtering protect networks.
- **Understand** how security methods can work together to provide greater protection.

> **Specification note.** AQA 8525 requires knowledge of exactly six protocols: TCP, IP, HTTP, HTTPS, SMTP, and IMAP. You must know what each protocol is used for. The TCP/IP model has four layers (not seven like the OSI model, which is not required for this specification).

Toolkit

Protocol purposes at a glance

Protocol	Full name	Purpose (exam answer)
TCP	Transmission Control Protocol	Sets up communication between hosts; ensures reliable data delivery; agrees packet size
IP	Internet Protocol	Addresses and packages data for transmission; routes packets across the network
HTTP	Hypertext Transfer Protocol	Transfers web pages between web browsers and web servers
HTTPS	Hypertext Transfer Protocol Secure	Encrypted version of HTTP for secure web transactions
SMTP	Simple Mail Transfer Protocol	Sends emails between mail servers

Protocol	Full name	Purpose (exam answer)
IMAP	Internet Message Access Protocol	Retrieves and stores emails on a server; allows access from multiple devices

The 4-layer TCP/IP model

Layer 4: Application	HTTP, HTTPS, SMTP, IMAP
Layer 3: Transport	TCP
Layer 2: Internet	IP
Layer 1: Link	Network hardware (NIC)

Data flow: When sending, data moves down through the layers (Application → Link). When receiving, data moves up through the layers (Link → Application).

TCP/IP layer functions

Layer	Function (exam answer)
Application	Where network applications operate, such as web browsers and email programs
Transport	Sets up communication between two hosts; agrees settings such as packet size
Internet	Addresses and packages data for transmission; routes packets across the network
Link	Where network hardware (NIC) and OS device drivers operate

Network security methods

Method	What it does	Example use
Authentication	Verifies the identity of a user or device	Username and password login
Encryption	Scrambles data so it cannot be read if intercepted	HTTPS encrypts web traffic

Method	What it does	Example use
Firewall	Monitors traffic and blocks/allows based on security rules	Blocking access to malicious websites
MAC address filtering	Allows/blocks devices based on their physical hardware address	Only allowing registered school devices to connect

1. Understanding network protocols

A **network protocol** is a set of rules that govern how data is transmitted across a network. Protocols define everything from how devices identify themselves to how errors are detected and corrected.

1.1 What is a protocol?

Think of a protocol like the rules of a language. Just as English has rules for grammar and sentence structure that allow people to understand each other, network protocols have rules that allow computers to understand each other.

> **Definition:** A network protocol is a set of rules that govern how data is transmitted across a network.

Protocols specify:

- **Data format:** How data should be structured and packaged
- **Addressing:** How devices identify themselves and find each other
- **Error handling:** How to detect and recover from transmission errors
- **Timing:** When and how fast data should be sent
- **Sequence:** The order of operations in communication

1.2 Why are protocols necessary?

Without standardised protocols, network communication would be impossible. Consider what would happen if:

- One computer sent data in a format another computer couldn't understand
- Devices had no agreed way to identify themselves
- There was no way to check if data arrived correctly
- Different manufacturers used completely different communication methods

Protocols solve these problems by providing a common standard that all devices and software must follow. When you visit a website, your browser and the web server both follow HTTP/HTTPS protocols, so they can communicate successfully regardless of who made the hardware or software.

1.3 Protocol layers

Rather than having one giant protocol that handles everything, network communication is divided into **layers**. Each layer has specific responsibilities and uses specific protocols. This layered approach has several advantages:

- **Simplicity:** Each layer only needs to handle its own tasks

- **Flexibility:** Layers can be updated independently
- **Interoperability:** Different implementations can work together if they follow the same protocols
- **Troubleshooting:** Problems can be isolated to specific layers

The **TCP/IP model** is the layered model used on the internet and most modern networks. It has four layers, each with specific protocols and functions.

1.4 The six protocols you must know

AQA requires you to know the purpose of exactly six protocols:

Core protocols

- **TCP** — reliable data delivery
- **IP** — addressing and routing

Application protocols

- **HTTP/HTTPS** — web pages
- **SMTP/IMAP** — email

Worked Example 1 – Defining a protocol

Question: Define the term network protocol. [1 mark]

Model answer:
A set of rules that govern how data is transmitted across a network **(1)**.

Exam tip: The key phrase is "set of rules". You must also mention data transmission or communication. Saying "rules for computers" alone is too vague.

Worked Example 2 – Why protocols are needed

Question: Explain why network protocols are necessary. [2 marks]

Model answer:
Protocols provide standardised rules that all devices must follow **(1)**, allowing devices from different manufacturers to communicate with each other successfully **(1)**.

Exam tip: Focus on standardisation and interoperability — the ability for different devices to work together.

Worked Example 3 — Protocol layers benefit

Question: State one advantage of using a layered approach to network protocols. [1 mark]

Model answer:
Each layer can be developed or updated independently without affecting other layers **(1)**.

Exam tip: Other valid answers: simplifies design by dividing tasks; makes troubleshooting easier; allows different implementations to work together.

Worked Example 4 — Identifying protocols

Question: State the protocol used for each of the following purposes:
(a) Sending emails between mail servers [1 mark]
(b) Secure web browsing [1 mark]
(c) Routing data packets across networks [1 mark]

Model answer:
(a) **SMTP** (Simple Mail Transfer Protocol)
(b) **HTTPS** (Hypertext Transfer Protocol Secure)
(c) **IP** (Internet Protocol)

Exam tip: Learn what each protocol does — questions often describe a purpose and ask you to name the protocol.

2. The six protocols in detail

This section covers each of the six protocols you must know for AQA GCSE Computer Science. For each protocol, you should understand its full name, purpose, and where it operates in the TCP/IP model.

2.1 TCP (Transmission Control Protocol)

TCP operates at the **Transport layer** and is responsible for reliable data transmission between devices.

Purpose: TCP sets up communication between two hosts and ensures reliable data delivery. The hosts agree on settings such as the size of packets.

TCP performs several important functions:

- **Connection establishment:** Sets up a connection between sender and receiver before data transfer begins
- **Segmentation:** Breaks large amounts of data into smaller packets

- **Reliability:** Checks that all packets arrive correctly; requests retransmission of lost packets
- **Ordering:** Ensures packets are reassembled in the correct sequence
- **Flow control:** Manages the rate of data transfer to prevent overwhelming the receiver

TCP establishes a connection before sending data and confirms delivery afterwards. This makes it reliable for transferring important data like web pages and emails.

2.2 IP (Internet Protocol)

IP operates at the **Internet layer** and is responsible for addressing and routing data across networks.

> **Purpose:** IP addresses and packages data for transmission. It routes packets across the network to reach their destination.

IP performs these key functions:

- **Addressing:** Every device on a network has a unique IP address that identifies it
- **Packaging:** Adds header information to data, including source and destination addresses
- **Routing:** Determines the path that packets take across the network to reach their destination
- **Fragmentation:** Splits packets if they are too large for the network

IP addresses allow data to be sent to the correct device. When you visit a website, IP ensures that your request reaches the correct server and that the response comes back to your device.

2.3 HTTP (Hypertext Transfer Protocol)

HTTP operates at the **Application layer** and is used for transferring web pages.

> **Purpose:** HTTP transfers web pages between web browsers and web servers.

When you type a web address into your browser:

1. Your browser sends an HTTP **request** to the web server
2. The server processes the request and sends back an HTTP **response**
3. The response contains the web page content, which your browser displays

HTTP is a **request-response** protocol — the client (browser) requests, and the server responds. However, HTTP transmits data in plain text, which means it is **not secure**. Anyone intercepting the traffic can read the content.

2.4 HTTPS (Hypertext Transfer Protocol Secure)

HTTPS operates at the **Application layer** and is the secure version of HTTP.

> **Purpose:** HTTPS provides an encrypted version of HTTP for more secure web transactions.

HTTPS works the same way as HTTP, but with one crucial difference — all data is **encrypted** before transmission:

- **Encryption:** Data is scrambled so it cannot be read if intercepted
- **Authentication:** Verifies that you are connecting to the genuine website
- **Data integrity:** Ensures data has not been modified during transmission

You can identify HTTPS websites by:

- The URL beginning with `https://` instead of `http://`
- A padlock icon in the browser address bar

HTTPS is essential for sensitive transactions such as online banking, shopping, and entering login details. Most modern websites now use HTTPS by default.

2.5 SMTP (Simple Mail Transfer Protocol)

SMTP operates at the **Application layer** and is used for sending emails.

> **Purpose:** SMTP sends emails between mail servers.

When you send an email:

1. Your email client uses SMTP to send the email to your mail server
2. Your mail server uses SMTP to transfer the email to the recipient's mail server
3. The email is stored on the recipient's server until they retrieve it

Important: SMTP is only used for **sending** emails, not for receiving them. To retrieve emails, a different protocol (IMAP) is used.

2.6 IMAP (Internet Message Access Protocol)

IMAP operates at the **Application layer** and is used for retrieving and managing emails.

> **Purpose:** IMAP retrieves and stores emails on a mail server, allowing users to view and manage emails from multiple devices.

Key features of IMAP:

- **Server-based storage:** Emails remain stored on the server
- **Multi-device access:** You can access the same emails from your phone, tablet, and computer
- **Synchronisation:** Actions (read, delete, move) are synchronised across all devices
- **Folder management:** You can organise emails into folders on the server

> #### SMTP vs IMAP — the key difference
>
> **SMTP** = **S**ending emails (think: S for Sending)
> **IMAP** = **I**nbox access and management (think: I for Inbox)
> Both protocols work together: SMTP delivers the email to the server, IMAP allows the recipient to access it.

Worked Example 5 — TCP purpose

Question: Describe the purpose of TCP (Transmission Control Protocol). [2 marks]

> **Model answer:**
> TCP sets up communication between two hosts/devices **(1)** and ensures reliable data delivery by checking packets arrive correctly/requesting retransmission of lost packets **(1)**.

Exam tip: Mention both connection setup AND reliable delivery for full marks. AQA also accepts "agrees settings such as packet size" as a valid point.

Worked Example 6 – IP purpose

Question: Describe the purpose of IP (Internet Protocol). [2 marks]

Model answer:
IP addresses and packages data for transmission **(1)** and routes packets across the network to their destination **(1)**.

Exam tip: The two key functions are addressing and routing. Don't confuse IP with TCP — IP handles addressing, TCP handles reliable delivery.

Worked Example 7 – HTTP vs HTTPS

Question: Explain why HTTPS should be used instead of HTTP for online banking. [2 marks]

Model answer:
HTTPS encrypts the data being transmitted **(1)**, so sensitive information such as passwords and account details cannot be read if intercepted **(1)**.

Exam tip: Always link the technical feature (encryption) to the practical benefit (data cannot be read if intercepted). Both parts are needed for full marks.

Worked Example 8 – SMTP and IMAP

Question: Describe the purpose of SMTP and IMAP. [2 marks]

Model answer:
SMTP: Sends emails between mail servers **(1)**
IMAP: Retrieves/stores emails on a server, allowing access from multiple devices **(1)**

Exam tip: This is a common exam question. Remember: SMTP = Sending, IMAP = Inbox management. Don't confuse them!

3. The TCP/IP model and network security

The **TCP/IP model** organises network communication into four layers. Understanding this model helps you see how different protocols work together and where security measures fit in.

3.1 The 4-layer TCP/IP model

The TCP/IP model has four layers, each with specific responsibilities:

Layer 4: Application	HTTP, HTTPS, SMTP, IMAP
Layer 3: Transport	TCP
Layer 2: Internet	IP
Layer 1: Link	Network hardware (NIC), device drivers

↓ **Sending:** Data flows DOWN through layers (Application → Link)
↑ **Receiving:** Data flows UP through layers (Link → Application)

Layer 4: Application layer

This is where network applications operate, such as web browsers and email programs. The application layer provides the interface between user software and the network.

- **Function:** Provides network services directly to user applications
- **Protocols:** HTTP, HTTPS, SMTP, IMAP
- **Examples:** Viewing a web page, sending an email, using a messaging app

Layer 3: Transport layer

This layer sets up communication between the two hosts (sender and receiver). The hosts agree on settings such as the size of packets and how to handle errors.

- **Function:** Establishes communication; ensures reliable data delivery
- **Protocol:** TCP
- **Responsibilities:** Breaking data into packets, error checking, flow control, packet reassembly

Layer 2: Internet layer

This layer handles addressing and routing. It packages data for transmission and determines the path that packets take across the network.

- **Function:** Addresses and packages data; routes packets across the network
- **Protocol:** IP
- **Responsibilities:** IP addressing, packet routing, determining best path

Layer 1: Link layer

This is where the network hardware operates. It includes the Network Interface Card (NIC) and the operating system device drivers that control it.

- **Function:** Handles physical transmission of data
- **Components:** Network Interface Card (NIC), device drivers
- **Responsibilities:** Physical addressing (MAC addresses), converting data to signals

3.2 How data flows through the layers

When you send data (e.g., request a web page), it flows **down** through the layers:

1. **Application:** Your browser creates an HTTP request
2. **Transport:** TCP breaks the request into packets and adds reliability information
3. **Internet:** IP adds addressing information and determines the route

4. **Link:** The NIC converts the data to signals and sends it over the network
When data is received, it flows **up** through the layers in reverse order until the application can use it.

3.3 Network security – why it matters

Networks face many security threats: hackers trying to gain unauthorised access, malware spreading between devices, and attackers intercepting sensitive data. Network security measures protect against these threats.

AQA requires you to understand four key security methods and how they can work together.

3.4 Authentication

Authentication verifies the identity of a user or device before allowing access to the network or its resources.

> **Definition:** Authentication is the process of verifying the identity of a user or device.

Common authentication methods include:

- **Username and password:** The most common method; user proves identity with credentials they know
- **PINs:** Short numeric codes used to verify identity

Authentication ensures that only authorised users can access the network and its resources. For example, when you log into a school computer, you enter your username and password to prove you are a registered user.

3.5 Encryption

Encryption scrambles data so that it cannot be read if intercepted during transmission.

> **Definition:** Encryption is the process of scrambling data (converting plaintext to ciphertext) so it cannot be read if intercepted.

How encryption works:

- **Plaintext:** The original, readable data
- **Encryption algorithm:** Mathematical process that scrambles the data
- **Ciphertext:** The scrambled, unreadable data
- **Decryption:** Reversing the process using a key to recover the original data

Examples of encryption in networks:

- **HTTPS:** Encrypts web traffic between browser and server
- **Wi-Fi encryption:** Encrypts wireless network traffic to prevent eavesdropping
- **Email encryption:** Protects email content during transmission

3.6 Firewall

A **firewall** is a network security device that monitors incoming and outgoing network traffic and decides whether to allow or block specific traffic based on a defined set of security rules.

> **Definition:** A firewall monitors network traffic and blocks or allows traffic based on security rules.

Firewalls can:

- **Block unwanted connections:** Prevent hackers from accessing your network
- **Filter content:** Block access to certain websites or services
- **Detect suspicious activity:** Identify and block unusual traffic patterns
- **Log activity:** Record network traffic for security monitoring

Firewalls can be **hardware** (dedicated devices) or **software** (programs running on computers). Many networks use both types for layered protection.

3.7 MAC address filtering

MAC address filtering controls which devices can access a network based on their unique physical hardware address.

> **Definition:** MAC address filtering allows or blocks devices based on their physical address embedded in the network adapter.

Every Network Interface Card (NIC) has a unique **MAC address** — a physical address burned into the hardware. MAC filtering works by:

- **Whitelist:** Only allowing devices with approved MAC addresses to connect
- **Blacklist:** Blocking specific MAC addresses from connecting

For example, a school might configure its wireless network to only allow registered school devices to connect, blocking personal devices that aren't on the approved list.

3.8 Security methods working together

Individual security methods have limitations, but when used together they provide much stronger protection. This is called **defence in depth** or **layered security**.

> #### Example: Securing a school network
>
> - **Firewall:** Blocks access to inappropriate websites and prevents external attacks
> - **Authentication:** Students and staff must log in with username and password
> - **Encryption:** Wireless traffic is encrypted using WPA2
> - **MAC filtering:** Only registered school devices can connect to the network
>
> An attacker would need to bypass all four measures to gain unauthorised access.

Worked Example 9 — TCP/IP layers

Question: Name the four layers of the TCP/IP model in order from top to bottom. [2 marks]

> **Model answer:**
> 1. Application layer
> 2. Transport layer
> 3. Internet layer
> 4. Link layer
>
> **(1 mark for two correct, 2 marks for all four correct in order)**

> **Exam tip:** Check which direction the question asks for! "Top to bottom" means Application first. "Bottom to top" means Link first.

Worked Example 10 – Protocol layers

Question: State which layer of the TCP/IP model each of the following protocols operates at:
(a) HTTP [1 mark]
(b) TCP [1 mark]
(c) IP [1 mark]

Model answer:
(a) Application layer
(b) Transport layer
(c) Internet layer

Exam tip: Learn the protocol-layer mapping: Application (HTTP, HTTPS, SMTP, IMAP), Transport (TCP), Internet (IP), Link (hardware).

Worked Example 11 – Firewall explanation

Question: Explain how a firewall helps to protect a network. [2 marks]

Model answer:
A firewall monitors incoming and outgoing network traffic **(1)** and blocks or allows traffic based on defined security rules **(1)**.

Exam tip: Both parts are needed: what the firewall does (monitors traffic) and how it makes decisions (security rules).

Worked Example 12 – Security methods together

Question: A company uses encryption and authentication on its network. Explain how these two methods work together to improve security. [3 marks]

Model answer:
Authentication verifies the identity of users before they can access the network **(1)**. Encryption scrambles data so it cannot be read if intercepted during transmission **(1)**. Together, they ensure that only authorised users can access the network AND that data remains private even if intercepted **(1)**.

Exam tip: Explain what EACH method does individually, then explain the COMBINED benefit. This structure works for any "working together" question.

Extended response practice. AQA exams include extended response questions on network security. Here is an example with a model banded answer.

Extended Response — Network security in context

Question: A hospital stores sensitive patient records on its computer network. The network manager is reviewing security measures.
Discuss how the hospital could use authentication, encryption, firewalls, and MAC address filtering to protect patient data. [6 marks]

Model answer (Level 3, 5—6 marks):
The hospital should implement multiple security measures to protect sensitive patient records.
Authentication should be used to verify the identity of staff before they can access the network. Each member of staff would need to enter a username and password to log in. This ensures that only authorised personnel can access patient records, preventing unauthorised access from visitors or members of the public.
Encryption should be used to scramble data when it is transmitted across the network or stored on servers. If patient records are intercepted during transmission, the data would be unreadable without the decryption key. This protects patient confidentiality even if data is stolen.
A **firewall** should monitor incoming and outgoing network traffic and block any suspicious or unauthorised connections based on security rules. This would help prevent external hackers from accessing the hospital network and would block malware from communicating with external servers.
MAC address filtering could be used to ensure only registered hospital devices can connect to the network. Each computer and device has a unique MAC address embedded in its network adapter, so the hospital could maintain a whitelist of approved devices. This would prevent staff or visitors from connecting personal devices that might contain malware.
These four methods work together to provide layered security: authentication controls who can access the network, encryption protects data in transit, the firewall blocks malicious traffic, and MAC filtering controls which devices can connect. This defence in depth approach means an attacker would need to bypass multiple security measures to access patient data.

Exam tip (6-mark questions): To reach Level 3 (5—6 marks), you must explain ALL FOUR security methods, link each to the specific context (hospital/patient data), and explain how they work TOGETHER. Generic explanations without context will only reach Level 1—2.

Worked examples summary

This page contains 12 worked examples plus one extended response, covering all the key skills for network protocols and the TCP/IP model. Use this table to find specific types of questions for revision.

Example	Topic	Question type	Marks
1	Protocol definition	Define	1
2	Why protocols needed	Explain	2
3	Layered approach benefit	State	1
4	Identifying protocols	State	3
5	TCP purpose	Describe	2
6	IP purpose	Describe	2
7	HTTP vs HTTPS	Explain	2
8	SMTP and IMAP	Describe	2
9	TCP/IP layers	Name/State	2
10	Protocol-layer mapping	State	3
11	Firewall explanation	Explain	2
12	Security methods together	Explain	3
Ext.	Network security in context	Discuss (extended response)	6

Quick reference

Key definitions (mark-scheme style)

Term	Definition
Network protocol	A set of rules that govern how data is transmitted across a network.
TCP	Transmission Control Protocol — sets up communication between hosts and ensures reliable data delivery.
IP	Internet Protocol — addresses and packages data; routes packets across the network.
HTTP	Hypertext Transfer Protocol — transfers web pages between browsers and servers.
HTTPS	Hypertext Transfer Protocol Secure — encrypted version of HTTP for secure web transactions.

Term	Definition
SMTP	Simple Mail Transfer Protocol — sends emails between mail servers.
IMAP	Internet Message Access Protocol — retrieves and stores emails on a server; allows multi-device access.
Authentication	Verifying the identity of a user or device.
Encryption	Scrambling data (plaintext to ciphertext) so it cannot be read if intercepted.
Firewall	Monitors network traffic and blocks/allows based on defined security rules.
MAC address filtering	Allowing/blocking devices based on their physical hardware address.

Protocol-layer mapping

Layer	Protocols	Function
Application	HTTP, HTTPS, SMTP, IMAP	Where network applications operate
Transport	TCP	Sets up communication; reliable delivery
Internet	IP	Addressing and routing packets
Link	—	Network hardware (NIC) and drivers

Security methods summary

What each method does

- **Authentication:** Verifies identity
- **Encryption:** Scrambles data
- **Firewall:** Filters traffic
- **MAC filtering:** Controls device access

What each method protects against

- **Authentication:** Unauthorised access
- **Encryption:** Data interception
- **Firewall:** Attacks and unwanted traffic
- **MAC filtering:** Unauthorised devices

Typical pitfalls

Pitfall 1 — Confusing TCP and IP.
Wrong: "TCP routes packets across the network."
Right: TCP ensures reliable delivery. IP handles addressing and routing. Different protocols, different jobs.

Pitfall 2 — Confusing SMTP and IMAP.
Wrong: "SMTP is for receiving emails."
Right: SMTP sends emails. IMAP retrieves/stores emails. Remember: S = Sending.

Pitfall 3 — Saying HTTP is secure.
Wrong: "HTTP encrypts web traffic."
Right: HTTP is NOT secure — data is sent in plain text. HTTPS is the encrypted version.

Pitfall 4 — Incomplete encryption answers.
Wrong: "Encryption encrypts the data."
Right: Encryption scrambles data (1) so it cannot be read if intercepted (2). Both parts needed.

Pitfall 5 — Wrong layer order.
Wrong: Random layer order.
Right: Application → Transport → Internet → Link (top to bottom). Check question direction!

Pitfall 6 — Vague firewall explanation.
Wrong: "A firewall blocks hackers."
Right: A firewall monitors traffic and blocks/allows based on security rules. Be specific.

Pitfall 7 — Forgetting MAC is physical.
Wrong: "MAC addresses can be easily changed."
Right: MAC addresses are embedded in the physical network adapter hardware.

Pitfall 8 — Missing the "why" in protocol answers.
Wrong: "HTTPS uses encryption."
Right: HTTPS uses encryption SO THAT sensitive data cannot be read if intercepted.

Pitfall 9 — Confusing TCP/IP with OSI.

Wrong: Describing seven layers.
Right: AQA uses the 4-layer TCP/IP model only. OSI (7 layers) is not required.

Pitfall 10 — Vague protocol definition.
Wrong: "Rules for computers."
Right: A set of rules that govern how data is transmitted across a network.

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